

MUSKRAT DAM, ON, CanadaWMO: **716780**Lat: **53.440N**Lon: **91.760W**Elev: **278**StdP: **98.03**Time Zone: **-6.00 (NAC)**Period: **06-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.2	-34.5	-41.3	0.1	-37.0	-38.5	0.1	-34.4	9.1	-15.5	8.2	-15.6	1.6	220	0.697

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	27.4	18.1	25.8	17.3	24.3	16.7	20.1	25.0	18.9	23.3	17.8	22.1	4.2	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	13.8	22.0	17.3	12.8	20.9	16.1	11.8	20.0	59.2	25.0	55.0	23.5	51.6	22.2	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	8.0	7.1	-41.3	30.7	2.4	1.8	-43.0	32.0	-44.4	33.0	-45.7	34.1	-47.5	35.4		
(5)			-41.5	22.0	2.3	2.1	-43.1	23.5	-44.4	24.7	-45.7	25.8	-47.4	27.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.9	-20.2	-19.6	-10.6	-2.0	6.7	14.2	17.5	15.7	10.2	2.8	-8.5	-18.0
	DBStd	14.92	7.78	7.12	7.89	7.34	5.61	4.24	3.43	3.55	4.91	4.80	7.50	8.27
	HDD10.0	4627	937	829	640	361	134	11	1	2	58	231	554	869
	HDD18.3	7078	1195	1062	898	610	361	136	57	96	248	483	804	1128
	CDD10.0	650	0	0	0	1	33	136	232	178	63	7	0	0
	CDD18.3	60	0	0	0	0	1	11	31	15	3	0	0	0
	CDH23.3	548	0	0	0	0	27	112	264	118	26	1	0	0
	CDH26.7	88	0	0	0	0	5	9	55	16	4	0	0	0
Wind	WSAvg	3.2	3.0	3.1	3.2	3.3	3.5	3.1	3.1	3.1	3.2	3.4	3.1	3.0
Precipitation	PrecAvg	559	21	17	24	25	49	66	93	84	68	50	37	24
	PrecMax	729	36	35	46	48	108	157	153	124	144	94	62	55
	PrecMin	455	7	6	5	8	19	30	39	43	19	13	16	5
	PrecStd	63	8	7	10	11	19	24	32	23	28	20	12	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.1	1.4	13.4	19.3	26.6	27.8	31.1	28.5	26.2	19.9	10.0	0.5
		MCWB	-1.0	-0.1	9.2	10.2	14.6	17.1	21.3	19.2	17.5	13.2	7.3	-0.3
	2%	DB	-2.0	-1.8	7.7	15.2	22.6	26.2	28.0	26.2	23.2	15.5	6.0	-1.9
		MCWB	-2.4	-3.1	3.2	7.7	13.1	16.5	19.2	18.4	16.5	11.1	3.8	-2.4
	5%	DB	-4.6	-5.1	4.0	12.0	19.4	24.6	26.1	24.5	20.8	12.8	3.2	-3.9
		MCWB	-5.2	-6.4	1.2	5.9	11.4	16.0	17.6	17.5	15.5	10.0	1.9	-4.3
	10%	DB	-7.9	-8.0	1.2	9.0	16.7	22.6	24.5	22.6	18.1	10.2	0.9	-6.1
		MCWB	-8.3	-9.1	-0.8	4.1	10.0	14.5	17.0	16.6	14.5	8.0	0.2	-6.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.8	0.6	9.1	10.6	17.2	20.6	22.9	20.7	19.2	14.3	7.6	-0.4
		MCDB	-0.2	1.0	14.0	16.9	24.4	24.7	28.6	26.1	23.4	17.8	10.0	0.1
	2%	WB	-2.4	-3.1	3.7	8.3	14.1	18.3	20.7	19.5	17.6	11.9	4.4	-2.3
		MCDB	-1.9	-1.8	6.9	14.7	20.8	22.5	25.7	24.1	21.6	14.7	5.5	-2.0
	5%	WB	-5.2	-6.1	1.3	6.0	12.5	17.0	19.3	18.5	16.0	9.9	2.0	-4.3
		MCDB	-4.6	-5.1	3.6	11.7	18.2	22.3	24.2	22.7	19.3	12.5	3.1	-3.9
	10%	WB	-8.3	-9.0	-0.5	4.0	10.7	15.8	18.1	17.4	14.5	7.9	0.2	-6.5
		MCDB	-7.9	-7.9	1.3	8.1	16.3	20.7	22.3	21.6	18.1	10.5	1.1	-6.1
Mean Daily Temperature Range	5% DB	MDBR	10.2	12.9	14.1	13.5	11.6	11.4	10.9	10.5	9.4	6.5	7.3	8.5
		MCDBR	10.4	13.1	14.7	16.3	16.5	13.7	13.3	13.0	12.6	11.5	8.5	7.1
		MCWBR	10.0	11.7	11.1	10.1	8.6	6.4	6.3	6.4	6.8	7.3	6.7	6.7
	5% WB	MCDBR	10.4	12.9	12.5	15.5	14.3	11.5	11.1	11.3	10.7	10.4	7.8	6.9
		MCWBR	10.0	11.7	9.6	9.7	7.7	6.2	6.1	6.6	6.6	7.1	6.4	6.5
Clear-Sky Solar Irradiance	taub	0.237	0.276	0.279	0.321	0.347	0.354	0.384	0.358	0.328	0.287	0.264	0.227	
	taud	2.280	2.199	2.292	2.369	2.432	2.448	2.397	2.468	2.518	2.589	2.404	2.284	
	Ebn at Noon	765	825	913	913	902	896	859	866	854	831	727	705	
	Edh at Noon	57	88	100	107	106	106	110	97	81	61	52	47	
All-Sky Solar Radiation	RadAvg	1.04	2.24	3.75	5.26	5.28	5.62	5.38	4.40	2.97	1.65	0.84	0.66	
	RadStd	0.11	0.14	0.28	0.40	0.47	0.56	0.48	0.37	0.39	0.18	0.09	0.10	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page